Saurabh Patil

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Education

VPKBIET, Baramati **2021-2025**
Bachelor of Engineering in Electronics and Telecommunication , SPPU (Pune, India) *CGPA : 8.79*

Technical Skills and Interests

Languages: **Languages**: Java, Python, SQL, JavaScript, HTML5, CSS3
Frameworks: Spring Boot, Spring MVC, Spring JPA (Hibernate), Thymeleaf, Bootstrap, React.js, Flask, Django
Databases: Oracle Database, MySQL
Tools & Practices: Git, Maven, REST APIs, CI/CD, DevOps Fundamentals, Power BI
Core Concept: Object-Oriented Programming, Data Structures & Algorithms, DBMS, MVC Architecture, Service-Oriented/Microservices Architecture
Coursework: Web Development, Cloud Computing, Cyber Security, Deep Learning
Soft Skills: Analytical Thinking, Problem-Solving, Team Leadership, Effective Communication, Adaptability

Work Experience

Cognizant Technology Solutions **Nov 2025 – Present**
Programmer Analyst - Java Full Stack Development

- Completed structured training in Java full-stack development: Spring Boot, Spring MVC, Hibernate/JPA, REST APIs, and microservices concepts
- Engineered RESTful APIs and integrated backend services with responsive front-end components, optimizing data flow and application performance.
- Implemented object-oriented design principles and DevOps practices (Git, build automation, CI/CD) to ensure clean, maintainable, and enterprise-aligned development workflows.

Orbittal Electromech Eng Projects Pvt. Ltd., Pune **June 2024 – July 2024**
Software Engineer Intern

- Built a Python audio-signal comparison system processing **500+ samples** with **95% accuracy** in frequency matching for industrial inspections.
- Integrated a real-time analysis interface using **Flask** and REST APIs, lowering manual inspection effort by **30%** while meeting quality standards.

Centre of Excellence supported by Bharat Forge Ltd **Jan 2024 – Feb 2024**
Machine Learning Intern

- Trained a YOLOv8-based weed plant detection model on **1,000+ labeled images**, achieving **92% mAP** and reducing crop monitoring time by **40%**.

Projects

Transportation Fleet Management System (TFMS)

- **Technologies:** Java (Spring MVC), Spring JPA/Hibernate, Oracle Database, Maven, Thymeleaf, Bootstrap
- Architected and developed a full-stack fleet management system enabling vehicle tracking, trip management, fuel monitoring, and maintenance scheduling through a modular MVC architecture.
- Designed optimized relational database schemas with complex foreign key mappings in Oracle and implemented dynamic dashboards using Thymeleaf to enhance operational visibility and decision-making.

Audio Comparison using Python

- **Technologies:** Python, Librosa, NumPy, FFT, HTML, CSS, JavaScript
- Designed an audio inspection system for **Toyota Motors, Canada** achieving **95% accuracy** on **500%+ samples** and lowering false positives by **30%**.
- Automated frequency validation workflows to support real-time quality inspection in automotive assembly lines.

Weed Plant Detection using Deep Learning

- **Technologies:** Python, YOLOv8, Deep Learning, Computer Vision, NumPy
- Implemented a real-time weed detection model with **92% mAP** on **1,000+ images**, achieving sub-**50 ms** inference and reducing manual monitoring efforts by **40%**.
- Optimized detection pipelines to enhance field-level scalability, model efficiency, and deployment readiness.